**Web3Edu**

**LM01**

# What is Blockchain?

## Blockchain Fundamentals

*Understand the mechanism before judging the technology.*

### Extended Reading

Version 1.0 | October 2026



# Blockchain Fundamentals

**Extended Reading**
**Web3Edu - LM01: What is Blockchain?**

Blockchain technology first became widely known through the publication of Satoshi Nakamoto's 2008 paper. The real identity of the person - or group - behind the name Satoshi Nakamoto remains unknown.

The paper introduced what became the first and best-known application associated with blockchain technology: an electronic transaction system based on a new form of digital currency, Bitcoin. This explains why blockchain became closely associated with cryptocurrency in the public perception, particularly during its early years.

Interestingly, the word blockchain itself does not appear in Nakamoto's paper. Instead, the terms block and chain are used separately. Over time, the two words became combined into the term blockchain, which came to describe the underlying technology supporting systems such as Bitcoin.

KEY IDEA

**Blockchain is not another name for cryptocurrency. Bitcoin is an application of blockchain technology, not the definition of blockchain itself.**

Beyond the introduction of a digital currency, Nakamoto's work proposed a complete peer-to-peer electronic payment system. At a high level, such a system brings together several important components:

- **A distributed peer-to-peer (P2P) network, in which participants communicate and interact directly rather than relying on a central trusted intermediary.** This differs from conventional financial systems, where institutions such as banks typically act as intermediaries between transacting parties.

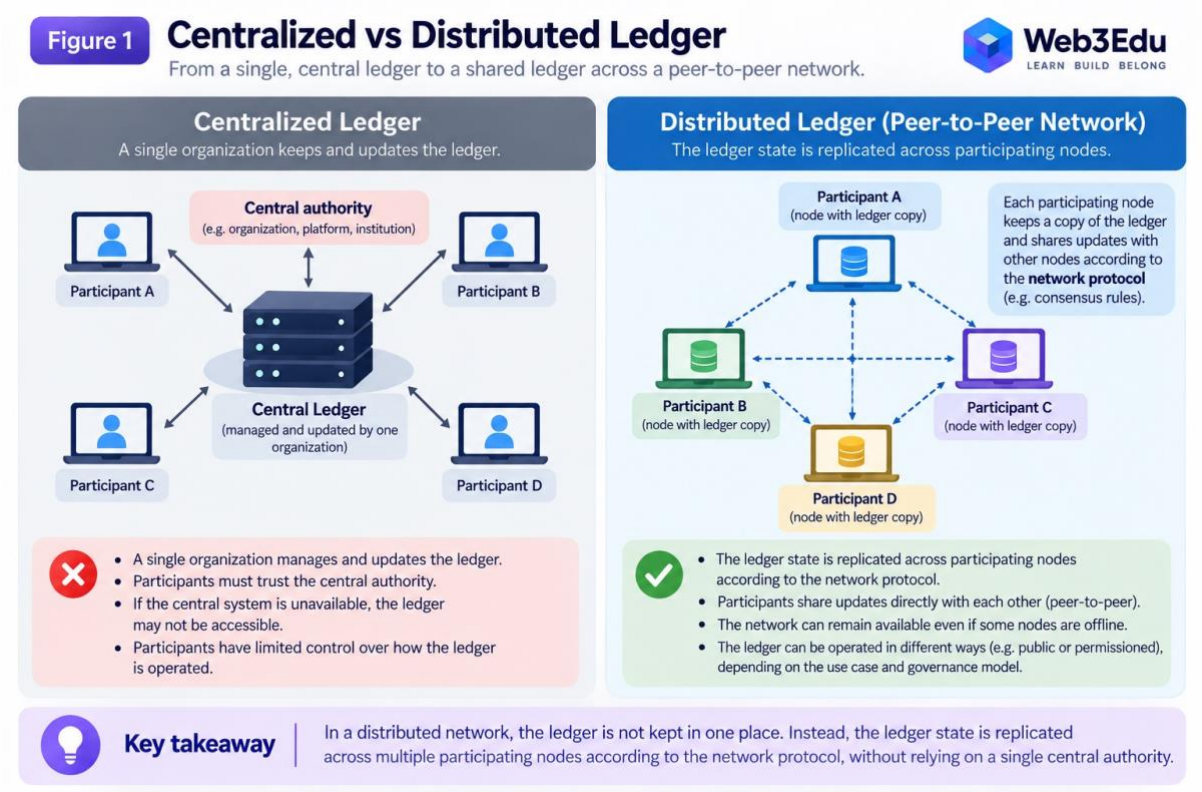



*Figure 1. Centralized vs distributed ledger. In a distributed ledger, ledger state is replicated across participating nodes according to the network protocol rather than being maintained only by a single central authority.*

- **A shared ledger that records transactions performed within the network.** Copies of this ledger are maintained across participating nodes and continuously updated according to the rules of the system. In a blockchain, the ledger is structured as a sequence of blocks containing transaction or other application data.
- Users interacting through wallets or similar interfaces, which allow them to manage the cryptographic keys required to interact with the blockchain network.

A blockchain-based system also typically relies on several additional mechanisms and properties:

- **Cryptographic keys and addresses.** Participants use cryptographic keys to authorize actions and interact with the network. Public identifiers or addresses can be derived from cryptographic key material and used when sending or receiving transactions. Private keys must remain under the control of their owners because they provide the ability to authorize operations associated with those keys.
- **Tamper-evident, append-oriented records.** Blockchain ledgers are generally designed so that new records are appended to an existing history rather than arbitrarily modifying previously accepted records. Cryptographic hashing and the linking of blocks make historical modification detectable and increasingly difficult under the assumptions of the particular blockchain protocol. This property should not be interpreted as an absolute guarantee that blockchain data can never be changed.

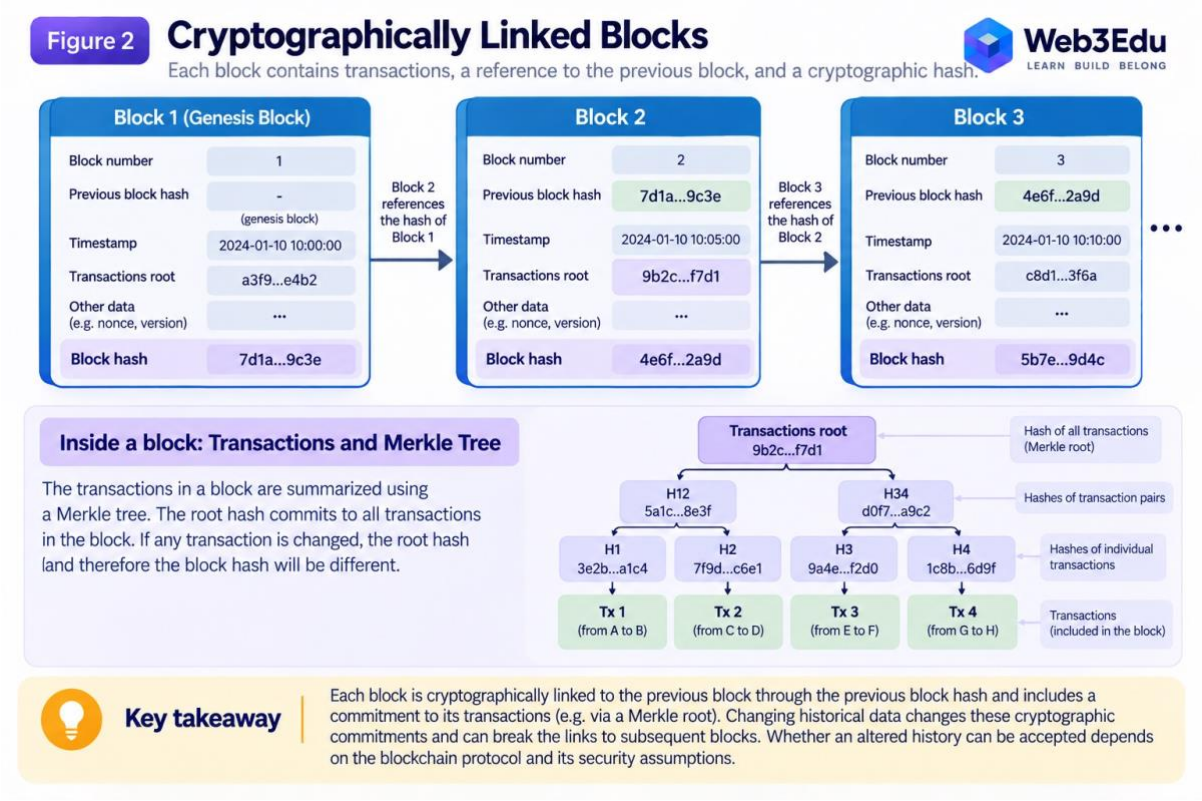



*Figure 2. Cryptographically linked blocks. Each block references information from the preceding block, so changes to historical data can affect the cryptographic links that follow.*

KEY IDEA

**Tamper-evident does not mean absolutely immutable. The strength of this property depends on the protocol and its operating assumptions.**

- **Chronological organization of transactions.** Transactions or other records are grouped into blocks, which are added sequentially to the existing chain. This provides an ordered history of accepted data and allows records to be associated with their position in the evolving blockchain.
- **Digital signatures.** Digital signatures are used to demonstrate that a transaction or operation was authorized using the appropriate private key. They therefore provide cryptographic evidence of authorization and help participants verify the authenticity of transactions without requiring a central authority.
- **Consensus mechanisms.** Because multiple nodes participate in maintaining the system, they require a mechanism for reaching agreement on the accepted state and history of the blockchain. Different blockchain systems employ different consensus mechanisms and assumptions. Well-known examples include Proof of Work (PoW) and Proof of Stake (PoS), which are explored later in the Web3Edu learning path.

KEY IDEA

**A distributed ledger is not automatically a decentralized system. Control also depends on governance, participation, consensus, and operating rules.**

## Putting It Together

Blockchain technology provides a way to organize records generated through interactions among participants in a distributed network. Data is grouped into blocks, and these blocks are cryptographically linked to form an ordered chain. The resulting ledger can be replicated among multiple participants, while cryptographic mechanisms, digital signatures, and consensus rules contribute different properties to the security and integrity of the overall system.

These mechanisms should not be considered guarantees in isolation. The properties achieved by a blockchain depend on its architecture, consensus protocol, governance model, implementation, and operating assumptions.

This distinction is important: blockchain is not simply another name for cryptocurrency, nor does the use of a blockchain automatically make a system decentralized, immutable, secure, or trustworthy. Those properties depend on how the system is designed and operated.

KEY IDEA

**Blockchain properties are outcomes of system design and operation - not guarantees produced simply by using a blockchain.**

## Before You Continue

Before moving to the next activity, make sure you can answer these questions in your own words:

- How do blocks and cryptographic hashes contribute to the structure of a blockchain?
- Why is tamper-evident a more accurate description than absolutely immutable?
- Why is blockchain not the same thing as cryptocurrency?
- Which parts of a blockchain system determine whether it is actually decentralized, secure, or trustworthy?

## Textbook Reading

This Web3Edu resource is original instructional material developed for LM01. It complements the course textbook and uses it as an academic reference; it is not presented as a translation of the textbook.

Recommended textbook reading: Patrikakis, C. Z., Leligou, E.-A., & Kogias, D. G. (2023). Αλυσίδες Συστοιχιών (Blockchain) - Εισαγωγή στην τεχνολογία και παραδείγματα. Kallipos Open Academic Editions. Chapter 1, especially §1.1, pp. 15-17.

ISBN: 978-618-5726-49-2 | DOI: 10.57713/kallipos-171

License: Original Web3Edu instructional material - © 2026 Dimitris Kogias · CC BY 4.0. The referenced textbook is licensed separately under CC BY-NC-SA 4.0.